

Ego-centric Video



User Overall Instruction (based on the task): Your user ID is cook_010, and today is May 9, 2026. First, you want to know what the next step is for the current dish and where the needed ingredients are. Then, ...



What's the next step for this dish? Things on the baking tray are ingredients of it.

Based on your actions and ingredients, you are on **step 2** of **Roasted Tomato Salmon**. The next step is cook salmon, which needs 200g salmon.



Okay. Where are they stored?

They are stored in the fridge.



Could you check if any other meat ingredients stored in the same place have expired? If so, add them to my shopping list.

Agent Overall Instruction: You are a service agent assisting users in **ego-centric video** environments. Your goal is to understand **user intents**, leverage **tools and context**, and complete requests with minimal back-and-forth. The tools available to you include ...



find_recipes_by_ingredients
("ingredient_name": ["tomato", "cornmeal"])

{"status": "success",
"message": "Roasted Tomato Salmon"}

get_cooking_steps
("recipe_name": "Roasted Tomato Salmon")

{"status": "success",
"message": "step1: roast tomatoes: put..."}



get_ingredient_location
("ingredient_name": "Salmon")

{"status": "success",
"message": "fridge"}

Database



Ingredients

[name, loc, nutr...]



Recipes

[name, steps, ingre...]



User Menus

[user_id, recipes...]



User Shopping Lists

[user_id, ingre...]

Environment



find_recipes_by_ingre

recipes = db.search(...)



add_to_shopping_list

db.add_item(...)



del_recipe_from_menu

db.delete_recpr(...)

Tool Library

Key
Aspects of
the Task

Multimodal Perception

Logical Reasoning

Associated Invisible Information Retrieval via Tools

Database State Modification via Tools

User-Agent
Dynamic
Interaction

User Response

Evaluated Agent Action

Agent-Tool
Dynamic
Invocation

Tool Invocation

Invocation Result